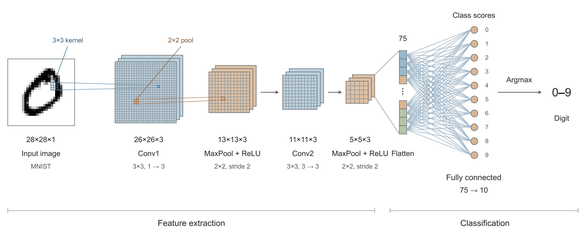
Sparsity-Aware Winograd Accelerator

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Introduction

CNN activations often contain zeros, a property known as activation sparsity, which allows unnecessary computations to be skipped. Winograd convolution reduces multiplications for small kernels by transforming inputs and filters, but its input transform does not necessarily preserve the original sparsity. We detect zero rows in the intermediate transform results and compact the execution schedule to reduce computation cycles, while integrated clock gating (ICG) suppresses unnecessary clock activity. We evaluate the individual effects of these techniques in a single core and extend the architecture to 16 cores with SRAM and multicast communication. Controlled experiments and VGG-16-BN workloads on CIFAR-10 examine when arithmetic savings translate into cycle and energy reductions.

1. MNIST Evaluation

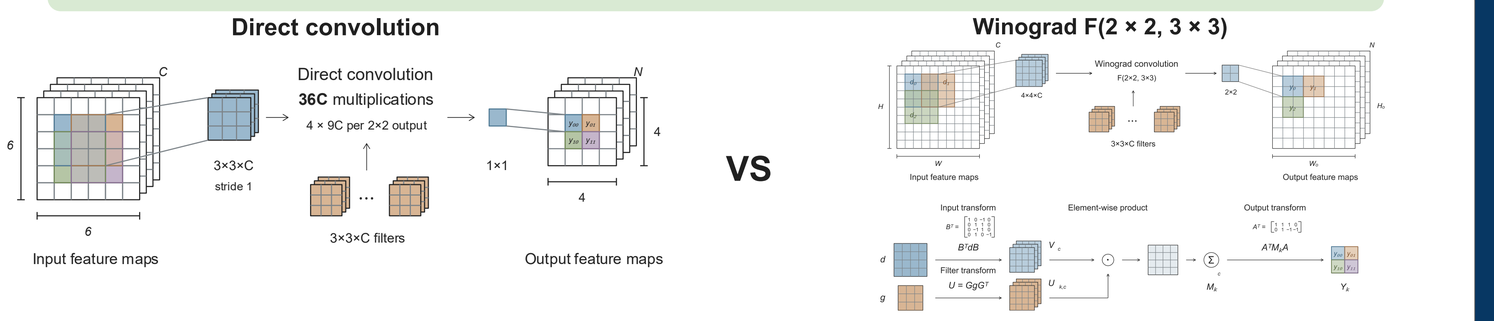


Task	Handwritten digit classification
Dataset	MNIST
Input	28 × 28 grayscale
Output	10 classes (0–9)
Accuracy	**97.0%**
Precision	INT4 W / UINT8 A

100 distinct images, repeated 10 times.

Zero-row skipping and ICG are compared under identical input and clock conditions.

2. Direct vs. Winograd

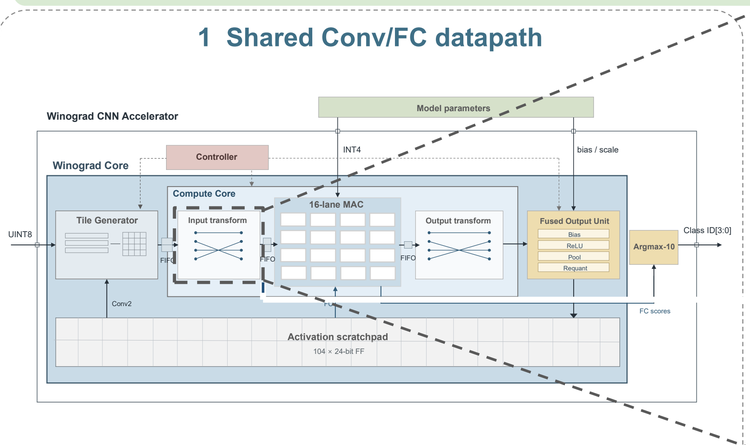


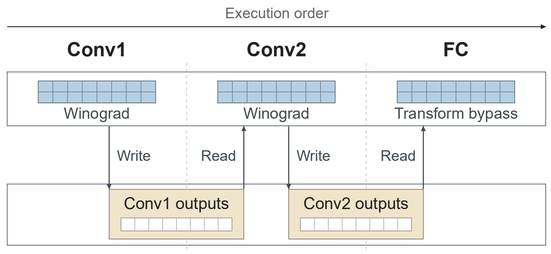
Same 2 × 2 output tile: 36 → 16 multiplications (−55.56%)

Compute neighboring outputs jointly through transforms, trading fewer multiplications for extra additions.
Weights are pre-transformed offline; activation tiles are transformed during inference.

3. Sparsity-Aware Core Architecture

1 Shared Conv/FC datapath





2 Input transform: matrix coefficients become adders and subtractors

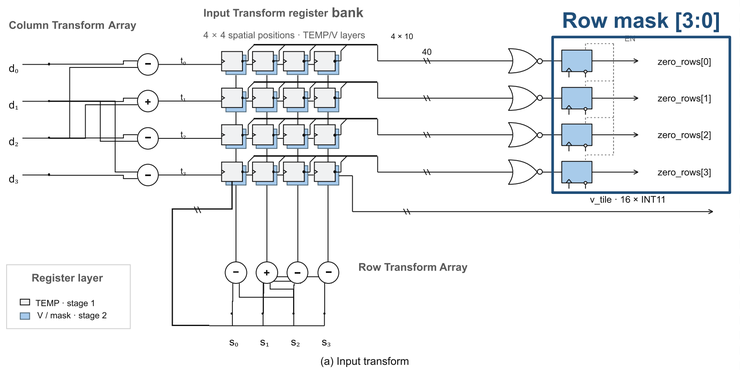
$$V = B^T d B$$

$$B^T = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

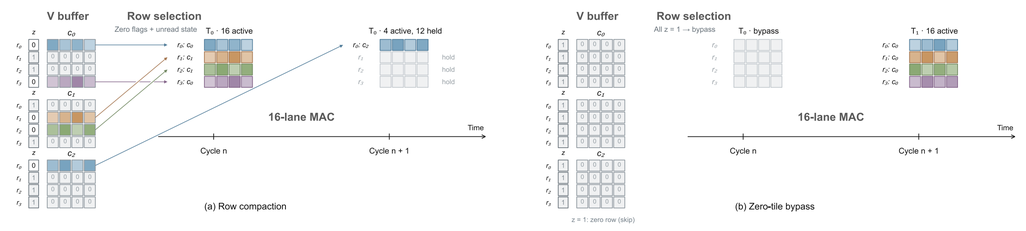
For each column

$$t_0 = d_0 - d_2$$
$$t_1 = d_1 + d_2$$
$$t_2 = d_2 - d_1$$
$$t_3 = d_1 - d_3$$

Row-wise NOR on T

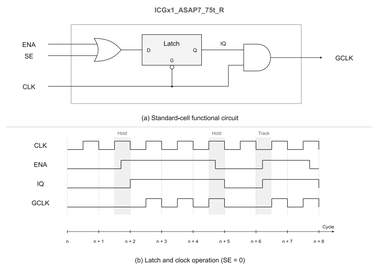


3 Zero-row skipping: compaction and tile bypass



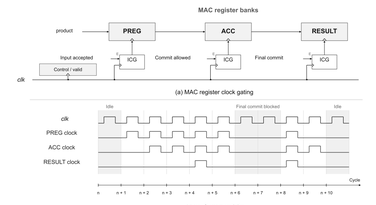
4. Clock Gating

Mapped to ASAP7 ICGx1_ASAP7_75t_R · 16 instances



Latch-based ICG suppresses unnecessary clock pulses without glitches.

The cell captures the enable signal during CLK-low and holds it during CLK-high.

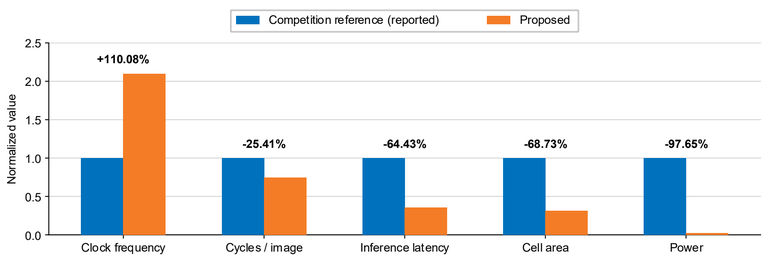


MAC register banks are clocked only when an update is required.

Separate enables control the product, accumulator, and result registers according to input acceptance, accumulation commit, and final-result commit.

5. Single-Core Results

Reported competition reference vs. proposed

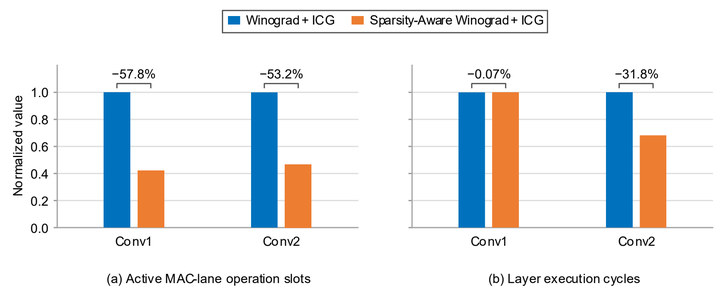


Dense-reference energy efficiency: 46.51 TOPS/W

$$\eta_{\text{ref}} = \frac{2N_{\text{MAC,ref}}f}{10^{12}P} = \frac{2 \times 183 \times 10^9}{10^{12} \times 0.007868472} = 46.51\ \text{TOPS/W}$$

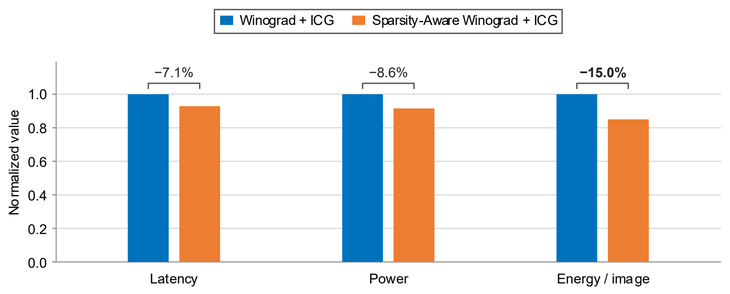
Calculated using 183 reference MACs at 1 GHz and 7.868 mW, with two operations per MAC.

Matched Winograd + ICG: sparsity OFF vs. ON



Zero-row skipping reduces Conv2 execution cycles by 31.8%.

MAC-lane operations decrease in both layers, while the external 1 pixel/cycle input limits Conv1 cycle reduction.

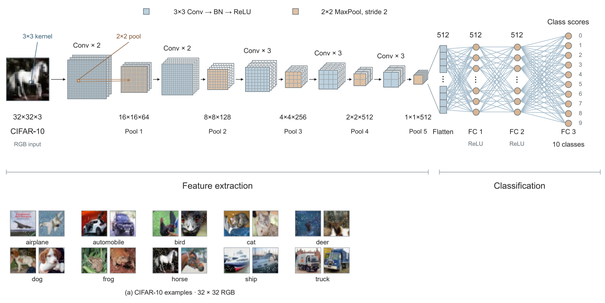


Zero-row skipping further reduces energy per image by 15.02%.

ICG is enabled in both cases, isolating the additional benefit of zero-row skipping.

6. Multicore Architecture

CIFAR-10 workload

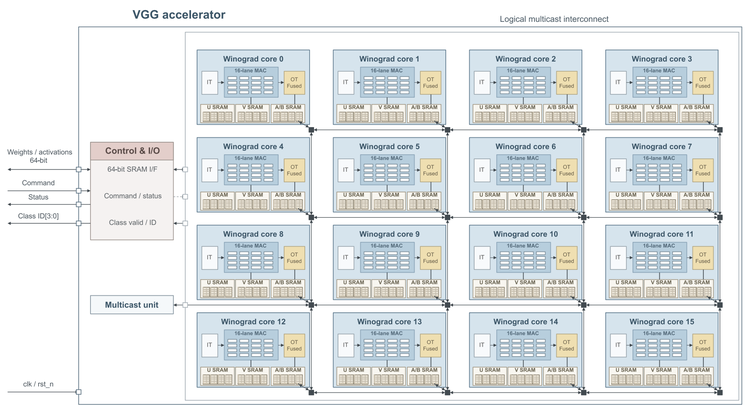


(a) CIFAR-10 examples · 32 × 32 RGB

VGG-16-BN: 13 Conv + 3 FC layers
CIFAR-10: 32 × 32 RGB, 10 classes

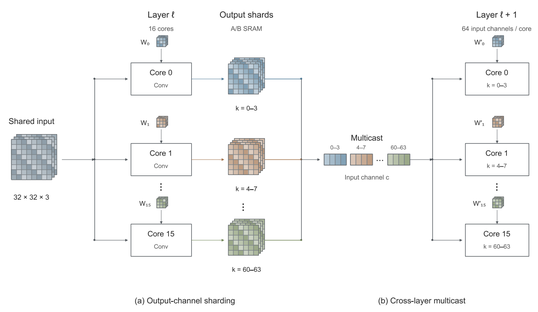
15,245,130 inference parameters (after BN folding)

The single-core architecture is replicated across 16 cores.



Each core retains Winograd, zero-row skipping, and ICG.

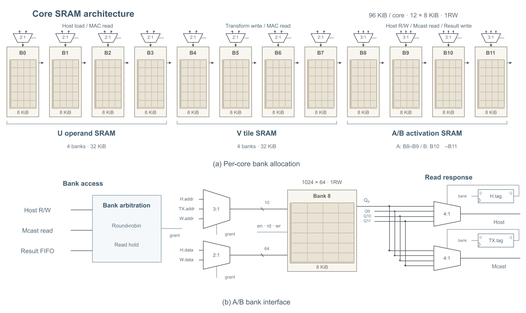
Output-channel sharding



Output channels are partitioned; activations are shared.

Each core computes a subset of output channels.
Multicast redistributes activations for the next layer.

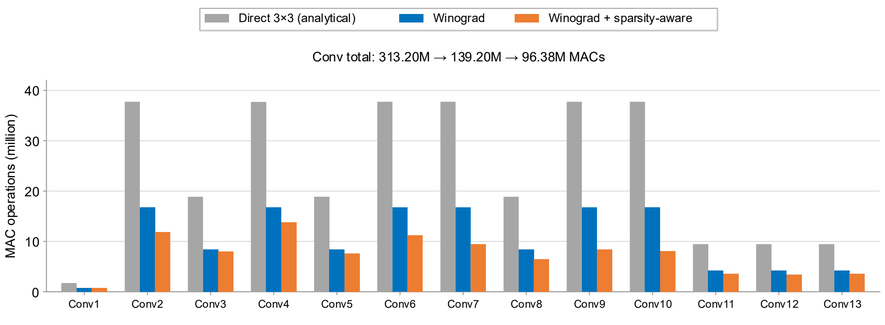
Per-core SRAM



Per-core SRAM supports reuse and intermediate storage.

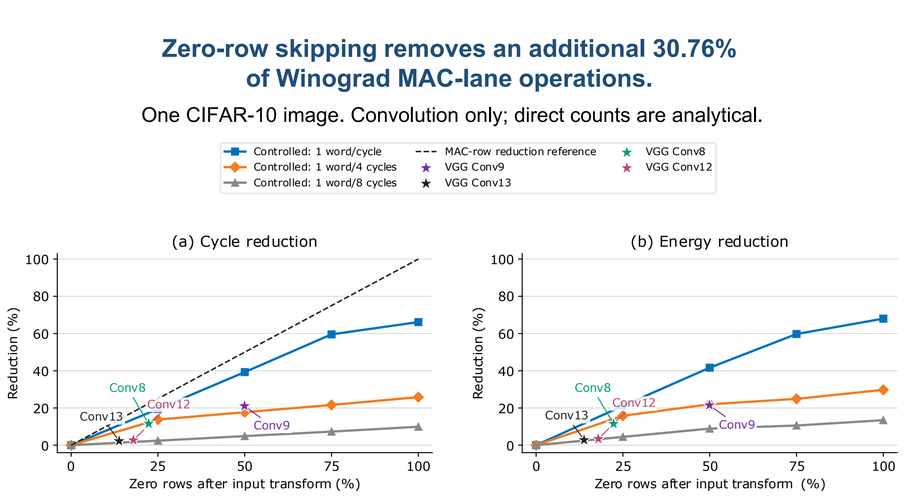
Separate banks store weights, transformed activations, and layer inputs and outputs.

7. Multi-Core Results



Zero-row skipping removes an additional 30.76% of Winograd MAC-lane operations.

One CIFAR-10 image. Convolution only; direct counts are analytical.



Controlled: 4×4×32 input, 8 output channels. VGG stars: one-image layer slices, 256 or 512 input / 4 output channels.
All weights resident; stars are application examples with different tensor shapes, not samples of the controlled curves.

Faster operand supply enables greater cycle and energy savings.

At 50% zero rows: **39.28%** fewer cycles and **41.64%** lower energy (1 word/cycle).
Core-level evaluation. Stars show measured VGG-16-BN layer examples.

ASAP7 predictive 7-nm library, post-synthesis evaluation. Single-core ablation: 1 GHz. Controlled core: 1.43 ns. github.com/jbpark0616/IDEC_Challenge